

## LAB 3: CIA Classification and Controls

**CIA = Confidentiality, Integrity, Availability**

| # | Scenario                                                                   | Affected CIA Property | Recommended Control                                        | Control Type           |
|---|----------------------------------------------------------------------------|-----------------------|------------------------------------------------------------|------------------------|
| 1 | A student's password is stolen and used to access private records.         | Confidentiality       | Multi-Factor Authentication (MFA)                          | Preventive             |
| 2 | An attacker reads confidential student records without permission.         | Confidentiality       | Access control / least privilege                           | Preventive             |
| 3 | An employee accidentally changes a student's grade.                        | Integrity             | Role-based access control + audit logs                     | Preventive / Detective |
| 4 | A hacker modifies university exam results.                                 | Integrity             | File integrity monitoring and access controls              | Detective / Preventive |
| 5 | A server crashes and students cannot access the university portal.         | Availability          | Server redundancy/failover                                 | Preventive             |
| 6 | Ransomware encrypts university files and makes them inaccessible.          | Availability          | Regular offline backups                                    | Corrective             |
| 7 | A staff member sends confidential student information to the wrong person. | Confidentiality       | Data-loss prevention (DLP) and security awareness training | Preventive             |
| 8 | A database record is changed without authorization.                        | Integrity             | Database access controls and audit logging                 | Preventive / Detective |
| 9 | A DDoS attack makes the university website unavailable.                    | Availability          | DDoS protection and traffic filtering                      | Preventive             |

| #  | Scenario                                                              | Affected CIA Property | Recommended Control                               | Control Type           |
|----|-----------------------------------------------------------------------|-----------------------|---------------------------------------------------|------------------------|
| 10 | An employee deletes important files by mistake.                       | Availability          | Regular backups and file recovery                 | Corrective             |
| 11 | Malware secretly changes system configuration files.                  | Integrity             | Endpoint protection and file-integrity monitoring | Detective / Corrective |
| 12 | Someone gains unauthorized access to a student's medical information. | Confidentiality       | Encryption and strict access permissions          | Preventive             |
| 13 | A power outage shuts down critical university servers.                | Availability          | UPS and backup generator                          | Preventive             |
| 14 | A malicious employee intentionally changes financial records.         | Integrity             | Separation of duties and audit logs               | Preventive / Detective |
| 15 | A lost laptop contains unencrypted confidential university data.      | Confidentiality       | Full-disk encryption                              | Preventive             |

## Student Questions

### 20. Can one incident affect all three CIA properties?

**Yes.** One incident can affect **Confidentiality, Integrity, and Availability simultaneously.**

For example, ransomware could:

- **Confidentiality:** steal and expose sensitive files.
- **Integrity:** modify or encrypt the original files.
- **Availability:** prevent users from accessing their data.

Therefore, a single security incident can have an impact on all three CIA properties.

### 21. Why is backup primarily an availability/recovery control?

Backup is primarily an **Availability and recovery control** because it provides another copy of data that can be restored when the original data is lost, deleted, corrupted, or encrypted by malware.

For example, if ransomware encrypts university files, a clean backup allows the university to **restore the files and resume operations**.

**Control classification:**

- **Preventive:** Stops or reduces an incident before it happens.
- **Detective:** Identifies or alerts on an incident.
- **Corrective:** Helps restore systems/data after an incident.